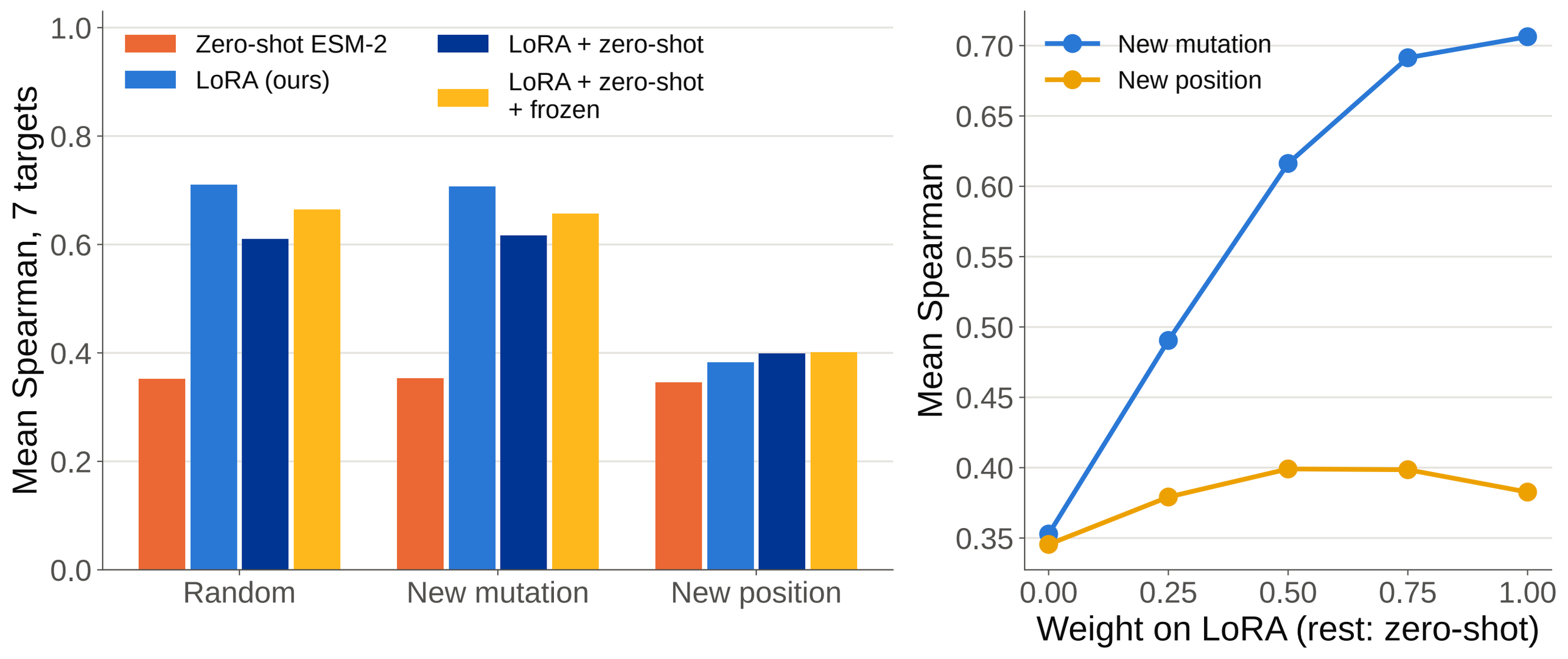
Zero-shot ESM-2
LoRA (ours)
LoRA + zero-shot
LoRA + zero-shot
+ frozen
Mean Spearman, 7 targets
1.0
0.8
0.6
0.4
0.2
0.0
Random
New mutation
New position
New mutation
New position
Mean Spearman
0.70
0.65
0.60
0.55
0.50
0.45
0.40
0.35
0.00
0.25
0.50
0.75
1.00
Weight on LoRA (rest: zero-shot)